MWE

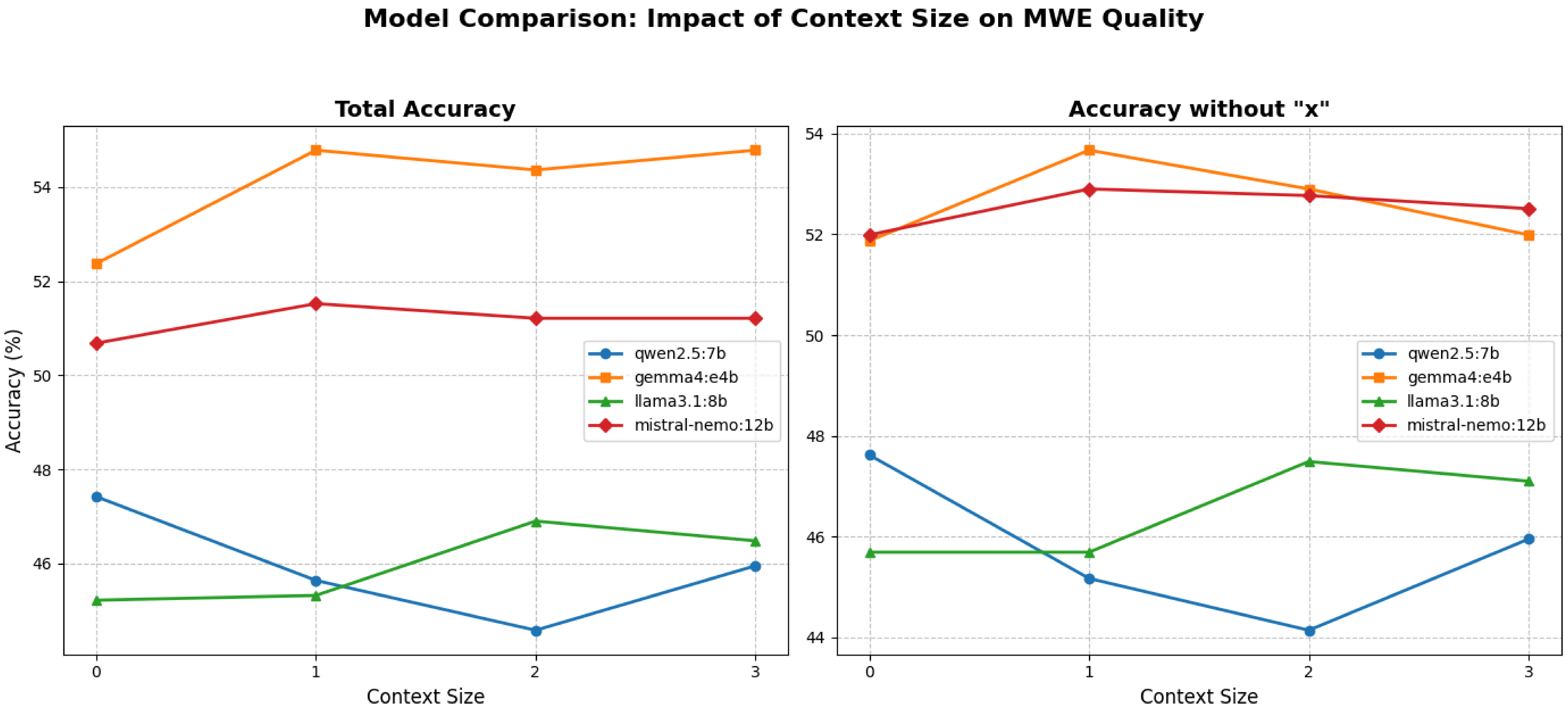
MULTI-WORD EXPRESSIONS

Vira Yurchenko

MULTI-WORD EXPRESSIONS (MWE)

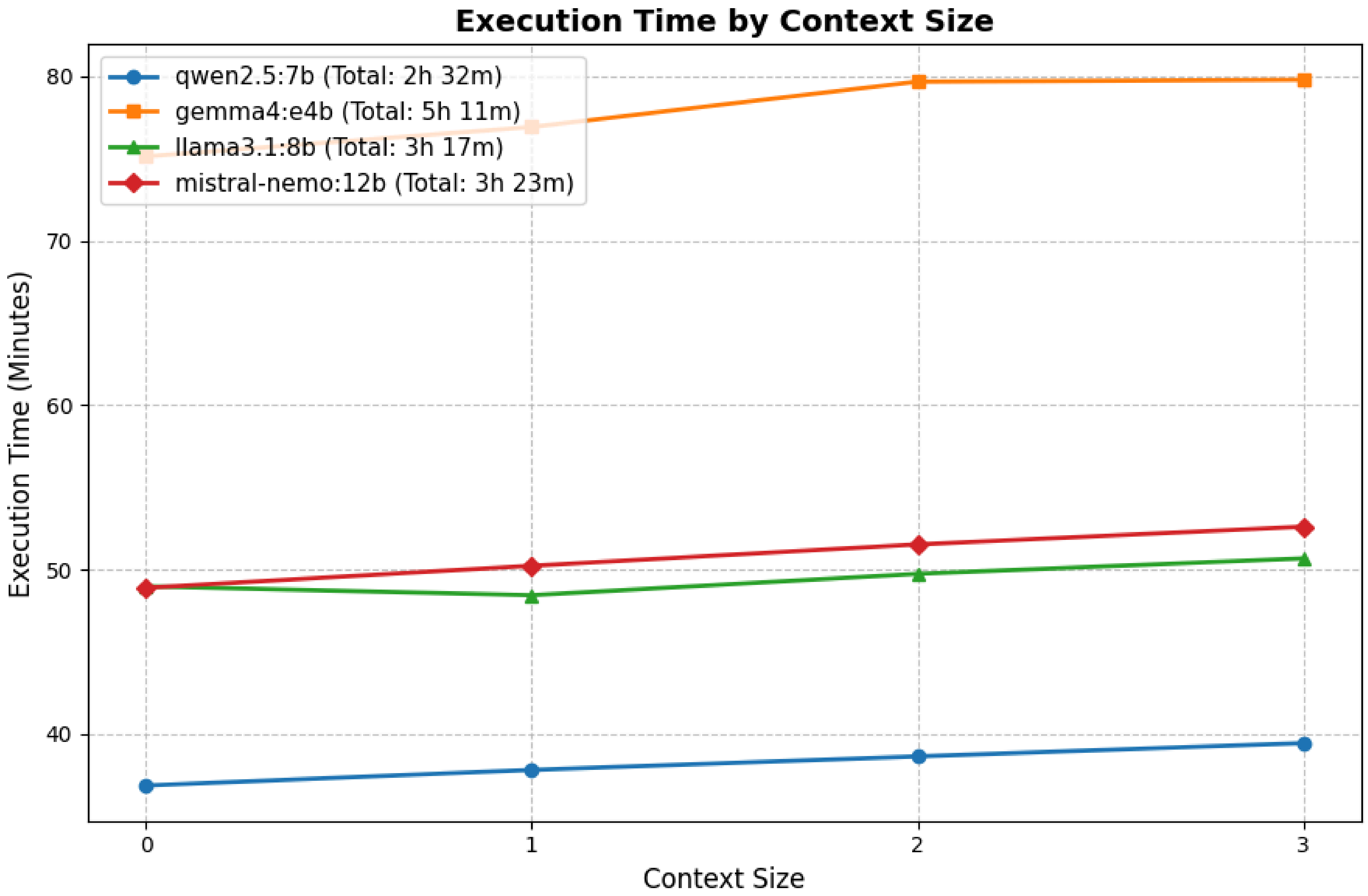
- The code processes an annotated JSON file containing already marked target words and expressions.
- The main challenge here is that phrases like “no longer” or “find out” have a unique meaning only when the words are together.
- The script finds a phrase in a file, it splits the phrase and extracts each dictionary definition for individual words.
- The model is then given this huge list.
- LLM analyzes the context, understands that the words belong together as one unit, and actively ignore all irrelevant meanings of separate words.

RESULTS



Model Comparison: Impact of Context Size on MWE Quality
Total Accuracy
Accuracy without "x"
Accuracy (%)
Context Size
qwen2.5:7b
gemma4:e4b
llama3.1:8b
mistral-nemo:12b

RESULTS

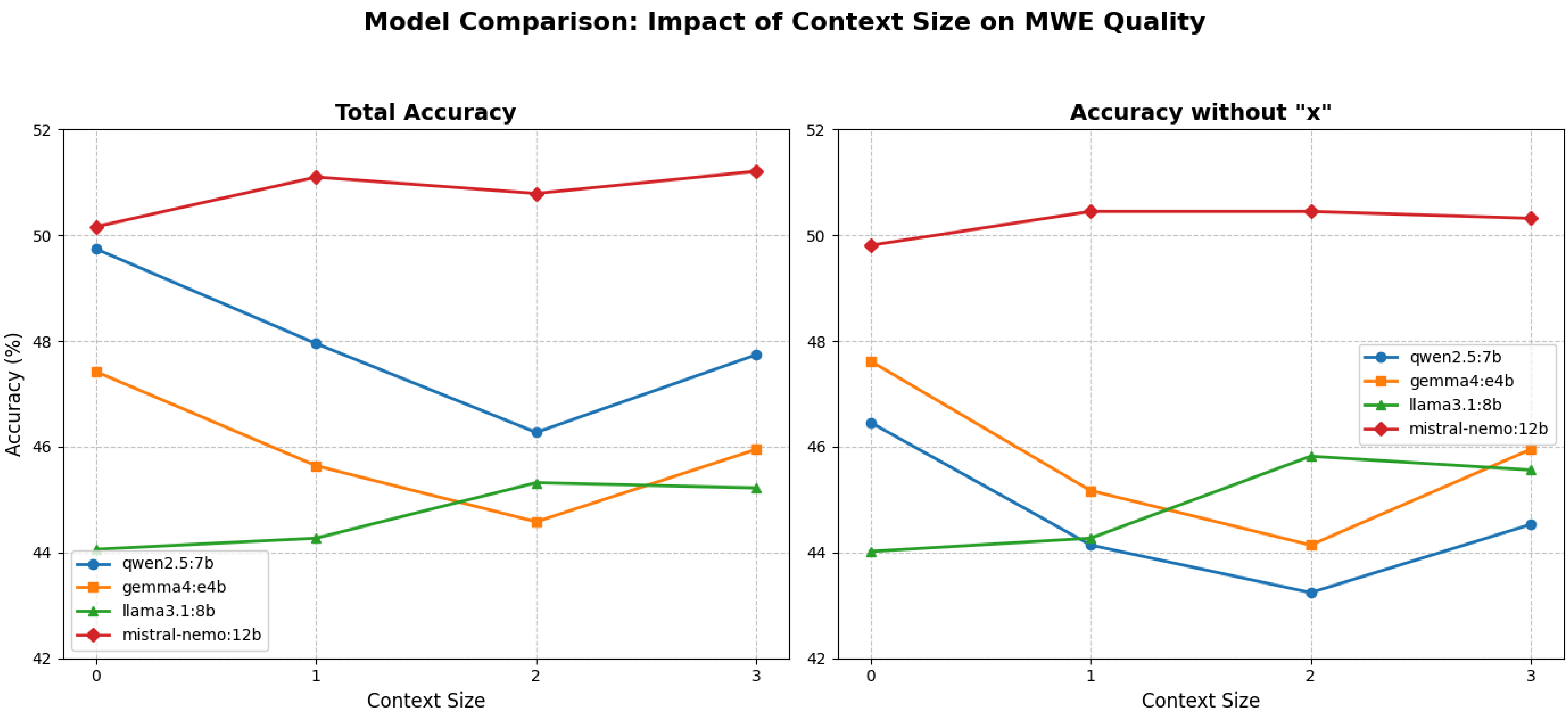


Execution Time by Context Size
qwen2.5:7b (Total: 2h 32m)
gemma4:e4b (Total: 5h 11m)
llama3.1:8b (Total: 3h 17m)
mistral-nemo:12b (Total: 3h 23m)
Execution Time (Minutes)
80
70
60
50
40
0
1
2
3
Context Size

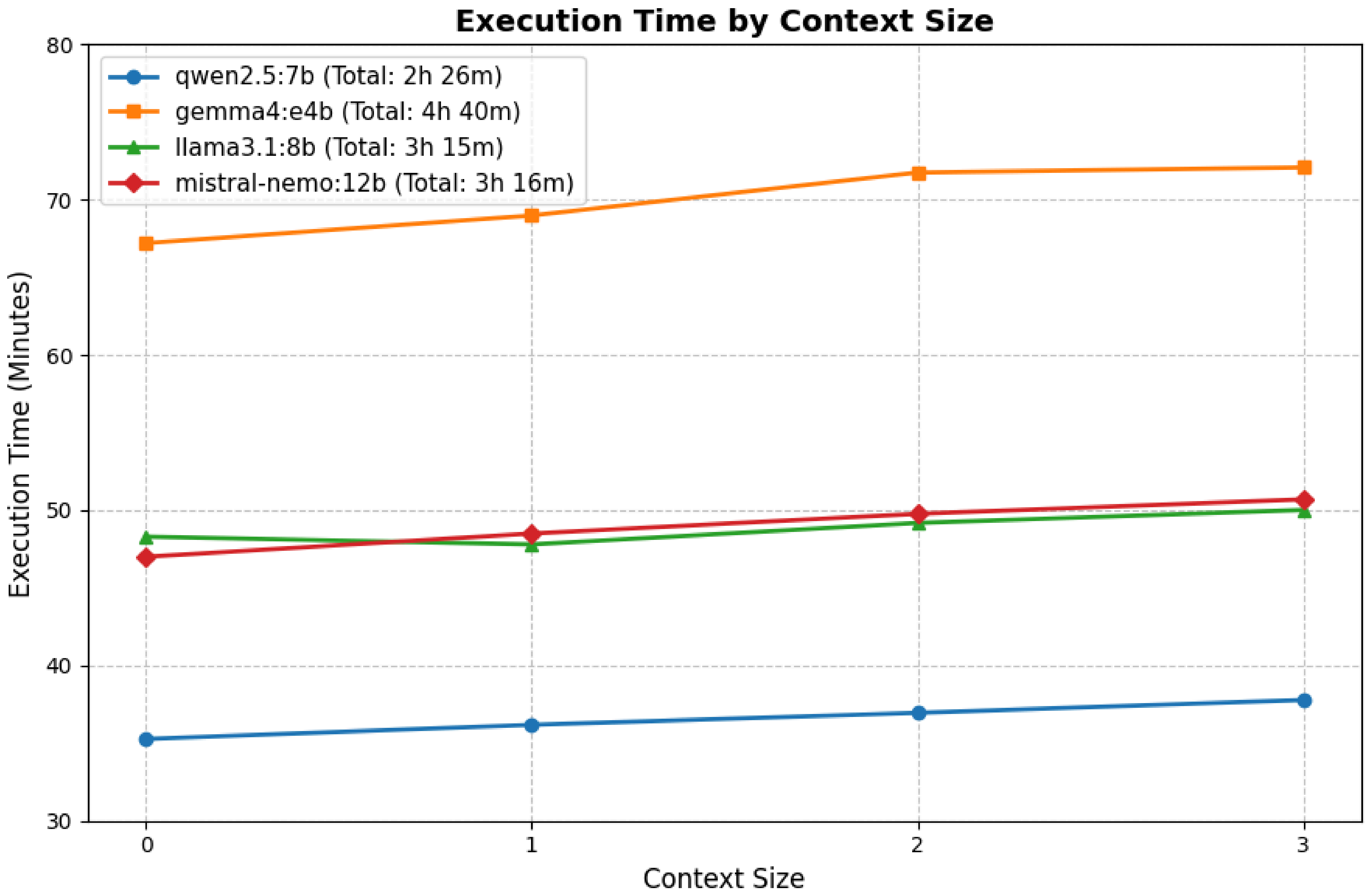
MULTI-WORD EXPRESSIONS (MWE)

- Alternative strategy, second option of processing.
- The script extracts the target multi-word expression from the input file.
- It requires WordNet only for the exact phrase ("find_out"), without splitting it into individual tokens.
- The LLM receives a highly targeted list of just 1–3 valid definitions, not plenty of irrelevant ones and selects the most accurate shade of meaning based on the context.

RESULTS


Model Comparison: Impact of Context Size on MWE Quality
Total Accuracy
Accuracy without "x"
Accuracy (%)
Context Size
Context Size
52
50
48
46
44
42
0
1
2
3
qwen2.5:7b
gemma4:e4b
llama3.1:8b
mistral-nemo:12b

RESULTS



Execution Time by Context Size
qwen2.5:7b (Total: 2h 26m)
gemma4:e4b (Total: 4h 40m)
llama3.1:8b (Total: 3h 15m)
mistral-nemo:12b (Total: 3h 16m)
Execution Time (Minutes)
30
40
50
60
70
80
0
1
2
3
Context Size

FUTURE WORK

- Show statistics for WSD with new model
- Process 100 sentences for MWE
- Write the report

THANK YOU FOR YOUR ATTENTION!